

Synthetic Financial Data Generation for Enhanced Financial Modeling

By

Christophe Dossou Hounwanou
christophe.hounwanou@aims.ac.rw

African Institute for Mathematical Sciences (AIMS), Rwanda

Supervised by: Dr. Yaé Ulrich Gaba

Department of Mathematics and Applied Mathematics

Sefako Makgatho Health Sciences University (SMU), Pretoria, South Africa

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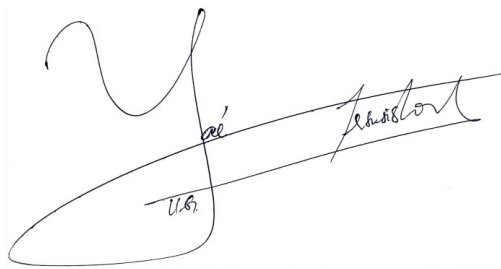
DECLARATION

This work was carried out at AIMS Rwanda in partial fulfilment of the requirements for a Master of Science Degree.

I hereby declare that except where due acknowledgement is made, this work has never been presented wholly or in part for the award of a degree at AIMS Rwanda or any other University.

A handwritten signature in black ink, featuring a large, stylized 'C' and 'D' that loop around each other, with the name 'Hounwanou' written below.

Student: Christophe Dossou Hounwanou

A handwritten signature in black ink, featuring a large, stylized 'Y' and 'U' that loop around each other, with the name 'Gaba' written below.

Supervisor: Prof. Dr. Yaé Ulrich Gaba

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Sincere thanks is extended to my parents in particular for their unwavering support and generosity, which have allowed me to continue my education.

For its kind support and for empowering African youth through leadership development and education, I am extremely grateful of the Mastercard Foundation.

DEDICATION

I dedicate this book above all to God Almighty, whose guidance and grace have sustained me throughout this journey.

My late father, Marius, whose ideals and goals continue to inspire me every day, is also honored in this thesis. For my beloved mother, Rosalie, and my siblings, for their unwavering love, prayers, and support.

Additionally, I would like to express my gratitude to my supervisor, Prof. Dr. Yaé U. Gaba, and tutor, Mr. Pierre, for their unwavering guidance, patience, and support throughout this endeavor.

Abstract

In the modern decision-making environment, financial data is a key component of all aspects of decision support, ranging from risk analysis models to portfolio optimization and fraud detection. However, available financial datasets are often missing due to prohibitive privacy restrictions, costs to access data, and/or the sensitiveness of financial data. The sparsity of accessible data is hindering research and innovation in developing financial models, because of the lack available and appropriate data for modeling. This thesis is concerned with utilizing machine learning approaches to create synthetic financial datasets, which maintain the statistical properties and statistical time dependencies of real-world financial datasets. The focus in this study with respect to synthetic data generation is time series financial datasets, transactional financial datasets, and portfolio datasets. Generative Models particularly, Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), are examined to generate realistic and diverse financial datasets. The synthesized datasets are evaluated through case studies, where case-specific financial modelling tasks, such as forecasting, risk evaluation and scenario analyses. While there has been technological implementation and evaluation of synthetic financial datasets, ethical and regulatory factors related to the use of synthetic data have been addressed, specifically with respect to data sharing for experimentation and innovation with respect to financial models, while acknowledging potential privacy constraints. The outcomes of this research support the creation of more innovative and resilient financial models by advancing approaches meant to alleviate the lack of data in this sector.

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1. Introduction

1.1 Motivation

Decision-making in sectors like banking, insurance, and investment management focuses on financial data. Risk assessment (Crouhy et al., 2000), portfolio optimization (Markowitz, 1952), fraud detection (Bolton and Hand, 2002), and algorithmic trading (Chan, 2013) are just a few of the essential applications it powers. As an example, investment firms use financial time series to forecast market behavior and optimize asset allocation strategies, while banks use historical transaction data to assess creditworthiness.

Accessing real-world financial data is still very difficult, despite its crucial role. Data availability is severely restricted by proprietary restrictions and strict privacy regulations like the General Data Protection Regulation (GDPR) (European Union, 2016). Further impediments are the high expenses of obtaining high-quality datasets, especially in academic research and innovation settings where resources might be scarce (Jordon et al., 2019).

A promising substitute for these limitations is synthetic financial data. In order to enable safe data sharing without violating privacy or proprietary rights, synthetic data creates datasets that closely mimic the statistical characteristics and temporal structures of actual financial data (Patki et al., 2016; Gonçalves et al., 2020). Additionally, it offers a scalable and reasonably priced platform for creating, evaluating, and validating financial models. The need to investigate sophisticated machine learning techniques that can produce realistic and practically applicable synthetic financial data for use in subsequent financial tasks is what motivated this dissertation.

1.2 Problem statement

Given the sensitive nature of financial transactions and privacy concerns governed by laws like GDPR (European Union, 2016), the availability of actual financial data is severely limited. These constraints make it difficult for scholars, developers, and organizations to obtain the high-quality datasets required for creating, honing, and verifying reliable financial models (Esteban et al., 2017; Wiese et al., 2020).

Moreover, these challenges are made worse by the dangers of disclosing private information and the fact that the majority of financial datasets are proprietary (Torkzadehmahani et al., 2019). Alternative data solutions that strike a balance between innovation and privacy protection are therefore urgently required.

Creating synthetic financial data that faithfully captures the distribution, structure, and temporal dependencies of actual datasets is one way to achieve this (Patki et al., 2016). In order to close this crucial gap, this dissertation examines efficient generative methods for creating realistic and private synthetic datasets for use in financial modeling applications.

1.3 Objectives

The main objective of this thesis is to develop and validate methods for generating synthetic financial data that are both realistic and suitable for practical financial modeling tasks.

1. To design generation techniques that simulate various types of financial data, including time series and transactional datasets (Yoon et al., 2019a).
2. To evaluate the effectiveness of the generated data through practical case studies in financial modeling and analysis (Wiese et al., 2020).
3. To examine the ethical and technical implications of using synthetic data within real-world financial applications (Gonçalves et al., 2020).

1.4 Contributions

This thesis makes the following key contributions to the field of synthetic financial data generation:

1. It proposes a modular framework for generating synthetic financial data, offering a step-by-step methodology applicable across multiple financial data types.
2. It explores the use of advanced generative models, including GANs, VAEs, and domain-specific architectures such as TimeGAN, tailored for financial time series (Yoon et al., 2019a; Wiese et al., 2020).
3. It presents a comprehensive evaluation pipeline with relevant metrics to assess the realism, utility, and privacy-preserving characteristics of the generated data (Esteban et al., 2017; Torkzadehmahani et al., 2019).

1.5 Structure of the dissertation

This dissertation is structured into six chapters, each progressively building upon the previous to provide a coherent exploration of synthetic financial data generation, from foundational concepts to practical implementation and evaluation. This dissertation is structured to provide a coherent progression from foundational concepts to practical implementation and analysis. Chapter 2 offers a comprehensive review of the fundamental principles underlying financial data, alongside an exploration of synthetic data generation techniques, with particular attention to ethical and practical considerations relevant to data privacy and regulatory compliance. Building on this foundation, Chapter 3 introduces the methodological framework, detailing the statistical approaches and machine learning technique such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) that are employed to generate realistic synthetic financial data. Following the methodological discussion, Chapter 4 focuses on the implementation process, describing

the tools, frameworks, datasets, and generative models utilized to operationalize the proposed techniques. The outcomes of these implementations are examined in Chapter 5, which presents the experimental results along with a critical analysis and discussion of the effectiveness, utility, and limitations of the generated synthetic datasets in various financial modeling scenarios. Finally, Chapter 6 concludes the dissertation by summarizing the key findings and contributions, while also identifying potential avenues for future research in the field of synthetic financial data generation and its application within financial analytics and decision-making contexts.

2. Background and literature review

2.1 Overview of financial Data

In a variety of industries, including banking, insurance, asset management, and corporate finance, financial data is essential for making decisions [Stadler et al. \(2022\)](#); [Chen and Ding \(2021\)](#). It includes comprehensive records about the operations, financial standing, and transactions of markets, entities, and instruments. Risk assessment, portfolio optimization, regulatory compliance, fraud detection, and algorithmic trading are among the crucial tasks that rely on this data [Chan \(2013\)](#); [Bolton and Hand \(2002\)](#).

A financial dataset can be represented as a time-indexed sequence:

$$D = \{x_t\}_{t=1}^T, \quad x_t \in \mathbb{R}^d$$

where x_t represents a feature vector at time t and T is the total number of time steps. When used in financial contexts, x_t may include one or more variables, including trading volume, the opening and closing prices, and derived features like log-returns. Sequences used in generative models such as TimeGAN or VAEs, as well as univariate and multivariate time series, may be modeled with this representation.

In general, there are basically three primary types of financial data, each with distinctive characteristics and uses:

Time Series Data: This group documents how financial variables, including commodity prices, interest rates, stock prices, and exchange rates, change over time. The characteristics of time series data, including autocorrelation, seasonality, volatility clustering, and non-stationarity, pose particular difficulties for forecasting and modeling [Box and Jenkins \(1970\)](#); [Bollerslev \(1986\)](#). For tasks such as risk management, market prediction, and derivative pricing, accurate modeling of such data is crucial [Tsay \(2010\)](#).

Transactional Data: Transactional datasets record individual financial operations, including bank transfers, credit card payments, trading activities, and other financial exchanges. These datasets are typically high-volume, high-velocity, and require robust data infrastructure for storage and real-time processing [Ngai et al. \(2011\)](#). Transactional data is vital for applications such as customer behavior analysis, anti-money laundering (AML), fraud detection, and credit scoring [Phua et al. \(2010\)](#).

Portfolio Data: Portfolio datasets reflect the composition and allocation of assets within investment portfolios. They include information on asset holdings, weights, returns, and associated risks. Such data is critical for portfolio performance evaluation, optimization strategies, and risk-return trade-off analysis, often leveraging modern portfolio theory and advanced quantitative models [Markowitz \(1952\)](#); [Fabozzi et al. \(2007\)](#).

Each of these financial data types introduces specific challenges. Financial time series, for instance, frequently exhibit heteroskedasticity, structural breaks, regime shifts, and heavy-tailed distributions, complicating traditional statistical modeling approaches [Bollerslev \(1986\)](#); [Hamilton \(1994\)](#). Transactional data, due to its scale and sensitivity, demands efficient data management systems and stringent security protocols to ensure privacy and compliance [Bellovin and et al. \(2019\)](#). On the other hand, portfolio data requires sophisticated techniques to model asset correlations, optimize diversification, and manage dynamic market conditions [Fabozzi et al. \(2007\)](#).

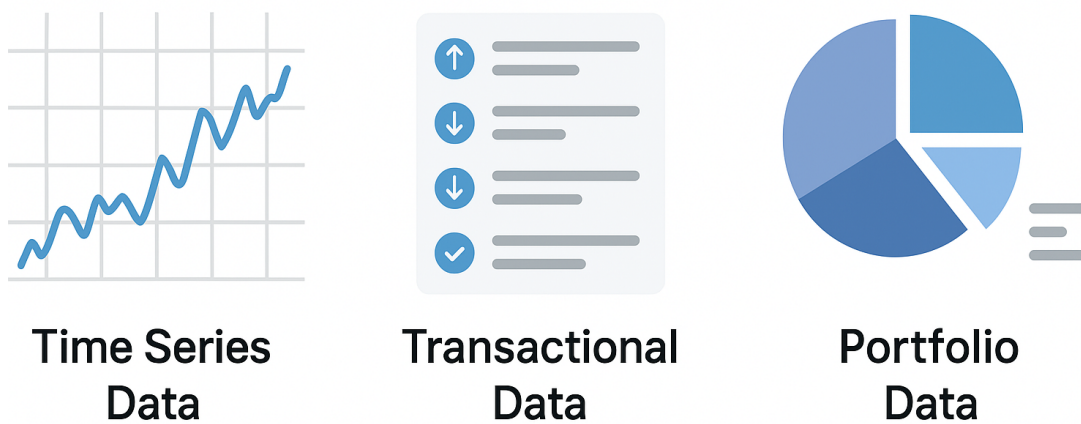


Figure 2.1: Illustration of different types of financial data: time series, transactional records, and portfolio snapshots.

Moreover, access to high-quality financial datasets is often constrained by strict privacy regulations, such as the General Data Protection Regulation (GDPR) [European Union \(2016\)](#), as well as proprietary restrictions imposed by financial institutions. The cost of acquiring comprehensive datasets further exacerbates these limitations, creating significant barriers for researchers and practitioners aiming to develop, test, and validate innovative financial models [Stadler et al. \(2022\)](#); [Gonçalves et al. \(2020\)](#).

With these challenges, synthetic financial data creation has been gaining more focus as a viable solution to facilitate research, experimentation, and model building without violating data privacy or at non-viable costs. The following sections will cover the principles, methods, and ethical considerations of synthetic data creation within the financial domain.

2.2 Synthetic Data Generation

Synthetic data refers to artificially generated data that closely resembles the statistical characteristics, interrelations, and structural patterns present in real data [Stadler et al. \(2022\)](#); [Gonçalves et al. \(2020\)](#). Synthetic data creation has recently been discovered to be an operational and tactical solution in the financial industry to address problems such as data scarcity, high procurement costs, and stringent privacy legislation, like those levied by the General Data Protection Regulation (GDPR) [European Union \(2016\)](#); [Bellovin and et al. \(2019\)](#).

The underlying idea of synthetic data generation is to produce datasets that are statistically equivalent to actual data without exposing any sensitive or personally identifiable information (PII) [Jordon et al. \(2019\)](#). It is achieved by training the actual data distribution $P_{\text{real}}(x)$ and sampling from an approximated distribution $P_{\text{synthetic}}(x)$ such that:

$$P_{\text{synthetic}}(x) \approx P_{\text{real}}(x)$$

By using this method, businesses can conduct thorough research, create predictive models, and thoroughly test financial systems without depending on private or proprietary data. Synthetic data encourages innovation while adhering to legal and ethical norms by separating data utility from privacy concerns.[\[Torkzadehmahani et al. \(2019\)\]](#).

Synthetic data has emerged as a promising solution for data sharing, augmentation, and privacy preservation across various domains. In the financial industry, generating realistic synthetic data holds immense potential for overcoming data silos, enhancing model development, and enabling secure collaboration.

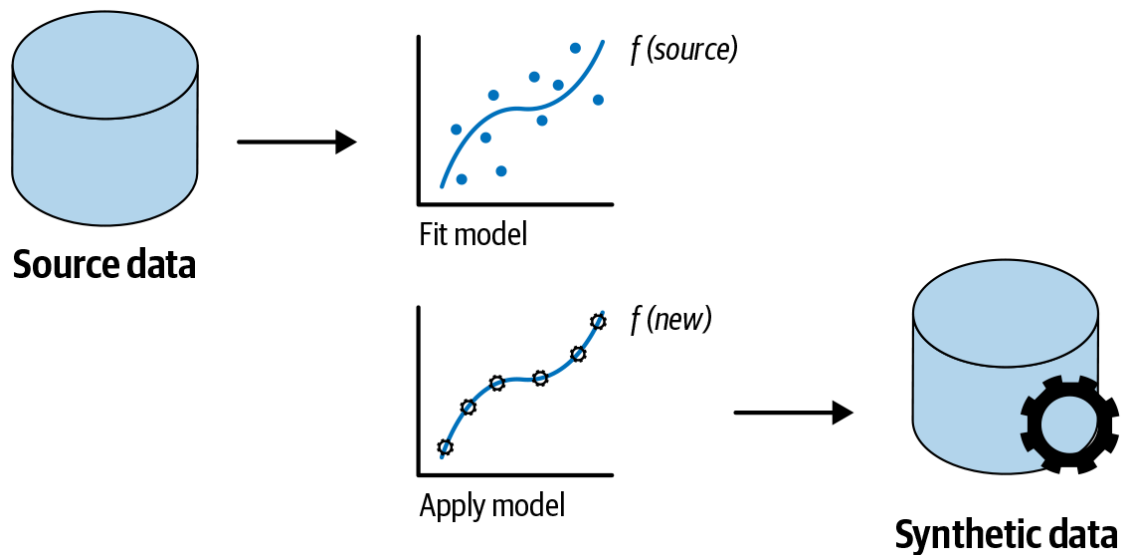


Figure 2.2: Generating Synthetic Financial Data

Synthetic data generation in the banking sector includes a variety of data types and applications:

Time Series Generation: To preserve important features like distributional properties, auto-correlation structures, seasonality, and volatility clustering built into real financial assets, such as stock prices, interest rates, and exchange rates [Yoon et al. \(2019b\)](#), artificial financial time series $\{x_t\}_{t=1}^T$ are created.

GAN-based models (e.g., TimeGAN) attempt to learn the joint distribution:

$$P(x_1, x_2, \dots, x_T) \quad \text{subject to temporal coherence}$$

These synthetic series are invaluable for developing and testing forecasting models, trading algorithms, and risk management frameworks.

Transactional Data Simulation: Synthetic transactional datasets replicate the patterns of financial activities, such as the volume, frequency, timing, and relational structures of customer transactions [Stadler et al. \(2022\)](#); [Ngai et al. \(2011\)](#). It can be represented as a set of events $\{e_i = (t_i, s_i, r_i, a_i)\}$, where each transaction includes a timestamp t_i , sender s_i , receiver r_i , and amount a_i . These datasets are critical for advancing fraud detection systems, customer behavior analysis, and credit risk modeling, especially when real transaction data is inaccessible due to privacy concerns.

Portfolio Data Generation: Weight vectors $w \in \mathbb{R}^d$, where $\sum_{i=1}^d w_i = 1$, are frequently used to model portfolios.

The goal of synthetic generation is to replicate realistic asset allocations and returns:

$$R_p = w^\top R, \quad \text{where } R \in \mathbb{R}^d \text{ is the asset return vector}$$

These artificial portfolios are employed in strategy validation and stress testing. [Fabozzi et al. \(2007\)](#).

The primary categories of synthetic data generation in finance are depicted in Figure 2.3, which also highlights the kinds of input data and common modeling pathways used for time series, transactional, and portfolio simulations.

Synthetic Data Generation Process

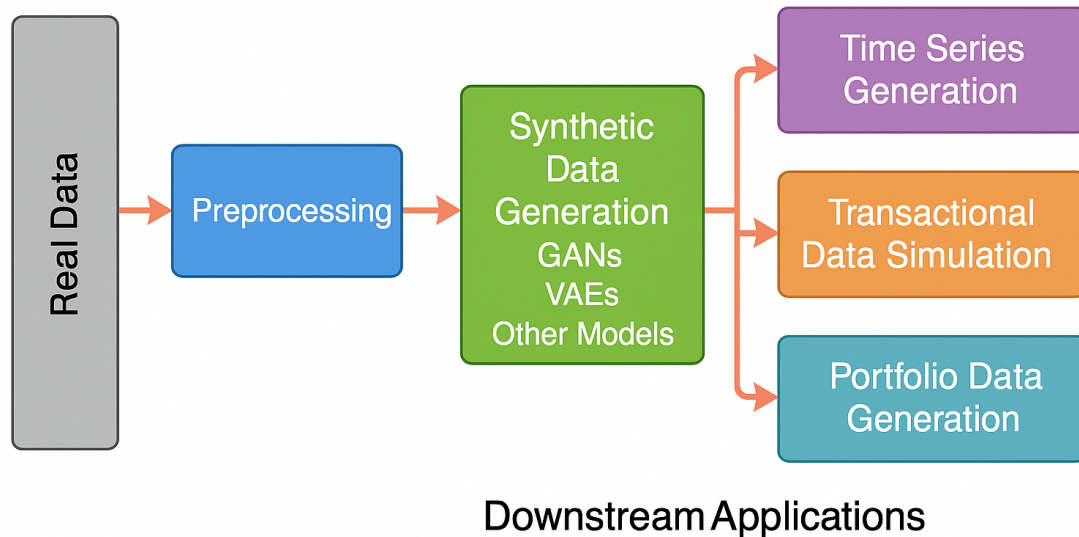


Figure 2.3: Illustration of synthetic data generation methods for time series, transactional, and portfolio data.

A number of important factors are driving the increasing use of synthetic financial data:

Privacy and Regulatory Compliance: Synthetic data enables financial institutions to exchange datasets for research, development, and collaboration without violating privacy laws or disclosing sensitive information [Bellovin and et al. \(2019\)](#); [Torkzadehmahani et al. \(2019\)](#).

Cost-Effectiveness: The generation of synthetic datasets reduces the time needed to gather enough real-world data for model training and validation [Gonçalves et al. \(2020\)](#) and does away with the need for expensive data acquisition procedures.

Risk Mitigation and Scenario Testing: Using synthetic data, financial models can be stress tested in uncommon but realistic market scenarios that may not be sufficiently reflected in historical datasets [Wiese et al. \(2020\)](#).

Although synthetic data offers multiple benefits there are a number of disadvantages as well. It is crucial to make sure that generated datasets preserve a high degree of fidelity to actual statistical distributions without overfitting to the original data [Esteban et al. \(2017\)](#). Additionally, caution must be exercised to avoid information leaks, especially when sensitive sources are used to create synthetic data [Jordon et al. \(2019\)](#). The advancement of synthetic data generation methods continues to revolve around striking a balance between privacy, effectiveness, and realism.

The following sections will examine the various approaches used to generate synthetic data, such as statistical methods and sophisticated machine learning models like Variational Autoencoders

(VAEs) and Generative Adversarial Networks (GANs), which have shown a great deal of promise in the financial industry.

2.3 Review of techniques for synthetic data generation

Many methods that combine traditional statistical models with modern machine learning techniques have been developed for the creation of artificial financial data. These methods aim to replicate the complex dynamics, relationships, and structures present in real-world financial datasets. This section summarizes the most significant approaches and provides information on their advantages, disadvantages, and possible uses.

2.3.1 Statistical Models. Because of their interpretability and mathematical rigor, traditional statistical methods have been used for a long time in financial time series modeling. Among these, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model [Bollerslev \(1986\)](#) and the Autoregressive Integrated Moving Average (ARIMA) model [Box and Jenkins \(1970\)](#) are particularly well-known.

In stationary time series data, ARIMA models are good at identifying autocorrelation patterns and linear trends. They do not, however, adequately address one of the key features of volatility clustering in financial data. To get around this, GARCH models were developed to simulate conditional heteroskedasticity, which is essential for generating realistic financial series that accurately represent market swings [Tsay \(2010\)](#).

Mathematically, an $ARIMA(p, d, q)$ model is expressed as:

$$\phi(B)(1 - B)^d x_t = \theta(B)\varepsilon_t$$

where B is the backshift operator, and ε_t is white noise. The differencing term $(1 - B)^d$ ensures stationarity when $d > 0$.

A $GARCH(p, q)$ model defines the conditional variance σ_t^2 as:

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

capturing volatility clustering observed in financial time series.

While these statistical approaches are computationally efficient and easy to interpret, they are limited in their ability to capture nonlinear relationships, complex dependencies, and structural breaks commonly observed in modern financial markets [Hamilton \(1994\)](#).

2.3.2 Deep Learning Models. The advent of deep learning has introduced powerful generative models capable of learning complex data distributions without explicit parametric assumptions.

Generative Adversarial Networks (GANs) [Goodfellow et al. \(2014\)](#) have revolutionized synthetic data generation by enabling the creation of highly realistic samples through adversarial training. A basic GAN consists of two neural networks: a generator G that produces fake data from noise $z \sim p_z(z)$, and a discriminator D that attempts to distinguish between real and fake samples. The training objective is formulated as a minimax game:

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{real}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

However, standard GAN architectures are primarily designed for independent and identically distributed (i.i.d.) data and struggle to model sequential dependencies inherent in financial time series.

To address this limitation, specialized architectures such as Time-series GAN (TimeGAN) [Yoon et al. \(2019b\)](#) have been proposed. TimeGAN integrates recurrent neural networks (RNNs) with adversarial learning to preserve both temporal dynamics and feature correlations, making it well-suited for generating synthetic financial sequences. Its objective combines three loss components:

- **Reconstruction Loss:** to preserve sequence structure using an autoencoder.
- **Adversarial Loss:** to ensure generated samples are indistinguishable from real sequences.
- **Supervised Loss:** to enforce temporal consistency between latent spaces across time steps.

Let $\mathbf{x}_{1:T}$ be a time series sample and $\hat{\mathbf{x}}_{1:T}$ its generated counterpart. The overall TimeGAN loss function becomes:

$$\mathcal{L}_{\text{TimeGAN}} = \mathcal{L}_{\text{recon}} + \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}$$

Variational Autoencoders (VAEs) [Kingma and Welling \(2013\)](#) offer another deep learning-based approach by learning probabilistic latent representations of data.

Given an input x , the encoder produces a latent distribution $q_\phi(z | x)$, while the decoder reconstructs data from sampled latent codes. The VAE optimizes a combination of reconstruction loss and Kullback-Leibler (KL) divergence:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - D_{\text{KL}}(q_\phi(z | x) \parallel p(z))$$

VAEs are effective in generating smooth and continuous synthetic datasets. However, they tend to oversmooth sharp transitions and abrupt changes, which are characteristic of financial market behaviors, potentially reducing the fidelity of generated data in high-volatility scenarios [Esteban et al. \(2017\)](#).

2.3.3 Summary of Techniques. A comparative overview of the techniques discussed can be found in Table 2.1, which highlights their main advantages and drawbacks in relation to the creation of synthetic financial data.

Table 2.1: Comparison of Synthetic Data Generation Techniques

Model	Main Strength	Limitation
ARIMA	Captures trends and autocorrelation	Ignores volatility clustering
GARCH	Models conditional volatility	Assumes parametric structure
GANs	Generates realistic samples	Challenges with sequential data modeling
TimeGAN	Preserves temporal dependencies	High training complexity
VAE	Learns smooth latent representations	Risk of oversmoothing outputs

2.3.4 Hybrid Methods. The interpretability of statistical models and the adaptability of deep learning techniques are combined in hybrid approaches that have been investigated recently [Bolton and Hand \(2002\)](#). For example, GANs or VAEs are used to model residual noise and complex stochastic patterns, while ARIMA models can be used to capture global trends and deterministic components. These hybrid frameworks improve the realism and robustness of synthetic datasets by utilizing the advantages of both paradigms. These techniques hold great promise for producing financial data that demonstrates both intrinsic market randomness and predictable structures.

As a result of their simplicity and transparency, statistical models offer a strong basis for the creation of synthetic data; however, deep learning models have greater capacity to capture complex, nonlinear dynamics. The drawbacks of each distinct technique are addressed by hybrid approaches, which offer a balanced route. The particulars of the financial data being modelled and the planned downstream applications play a major role in the method selection.

2.4 Practical and ethical considerations

There are significant ethical benefits to using synthetic financial data, especially when it comes to privacy protection and regulatory compliance. To comply with strict data protection frameworks like the General Data Protection Regulation (GDPR) [Bellovin and et al. \(2019\)](#); [European Union \(2016\)](#), institutions can foster research, innovation, and collaboration by creating datasets that replicate the statistical characteristics of real financial data without disclosing actual sensitive information. Safer data sharing across organizational and geographic boundaries is made possible by this approach, which reduces the legal, ethical, and reputational risks that are typically associated with handling real-world financial data [Gonçalves et al. \(2020\)](#).

In industries where confidentiality is crucial, synthetic data is a potent tool for removing obstacles to data accessibility. It enables academic researchers, fintech firms, and financial institutions to create, test, and validate machine learning models without violating contractual data agreements or violating client privacy [Stadler et al. \(2022\)](#). Additionally, by enabling contests, benchmarking, and cooperative projects that would otherwise be prohibited because of data sensitivity, synthetic datasets can promote open innovation.

Despite these benefits, there are also a number of moral and practical issues with the creation and application of synthetic financial data. A phenomenon known as membership inference attacks [Torkzadehmahani et al. \(2019\)](#) can occur when poorly designed generation processes

unintentionally encode recognizable patterns from the source data, potentially resulting in privacy leaks. Furthermore, biased models that perform poorly in real-world situations where such anomalies are significant may arise from synthetic datasets that miss important but infrequent market events, like financial crises or flash crashes. [Wiese et al. \(2020\)](#)

The potential risk of the model depending too much on artificial data is another issue. When used in real-world financial settings, models that have only been trained on artificial datasets might not be generalizable and produce disparities [Esteban et al. \(2017\)](#). This illustrates the importance it's to use synthetic data to supplement real-world data rather than to replace it.

Strict validation procedures are necessary to overcome these obstacles. To evaluate the statistical similarity between synthetic and real datasets, methods like Wasserstein Distance, Kolmogorov-Smirnov (KS) tests, and Maximum Mean Discrepancy (MMD) are frequently used [?]. Evaluations by domain experts are also essential for confirming that the synthetic data accurately depicts market behaviors and fails to leave out important financial trends.

The ethical use of synthetic financial data necessitates careful evaluation of data generation techniques, validation standards, and potential hazards, even though it presents a promising avenue for privacy-preserving innovation. For the financial sector to use synthetic datasets responsibly and effectively, it will be essential to conduct ongoing research on differential privacy, adversarial robustness, and fairness in synthetic data generation [Torkzadehmahani et al. \(2019\)](#).

3. Methodology

This chapter describes the methodological framework used to generate and evaluate synthetic financial data.

First, traditional statistical models are used to establish a baseline for the creation of synthetic data. Then, more intricate patterns and temporal dependencies are captured using contemporary machine learning techniques. Finally, the criteria by which the generated data was assessed for quality and realism are explained.

These techniques taken together guarantee that the synthetic data stays relevant for practical uses like portfolio optimization and risk management in addition to reflecting the statistical characteristics of actual financial series.

3.1 Statistical Techniques

First, synthetic financial time series were created using conventional statistical models. Generalized Autoregressive Conditional Heteroskedasticity (GARCH) and Autoregressive Integrated Moving Average (ARIMA) were the two primary methods used. An Augmented Dickey-Fuller (ADF) test was conducted as part of the methodical procedure to confirm the financial time series' stationarity.

By combining autoregression, differencing, and moving averages, ARIMA models are able to capture the linear dynamics of time series [Box and Jenkins \(1970\)](#). In order to model short-term dependencies and trends, an ARIMA(1,0,1) model was trained using historical S&P 500 data. The fit evaluation criteria and stationarity analysis were used to choose the model.

Studies have shown that prediction based on the **ARIMA** model often outperforms complex structural models [Yoon et al. \(2019a\)](#). The autoregressive model can be represented mathematically as:

$$x_t = \phi_0 + \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + \varepsilon_t - \Theta_1 \varepsilon_{t-1} - \Theta_2 \varepsilon_{t-2} - \cdots - \Theta_q \varepsilon_{t-q} \quad (3.1.1)$$

Where:

- x_t is the actual value at time t ,
- ε_t is the random error term at time t ,
- ϕ and Θ are the model coefficients,
- p and q are the orders of the autoregressive (AR) and moving average (MA) parts, respectively.

ARIMA, which stands for *Autoregressive Integrated Moving Average*, is characterized by three main parameters:

- p : the number of lag observations in the model (autoregressive terms),
- d : the degree of differencing required to make the series stationary (integration),
- q : the size of the moving average window.

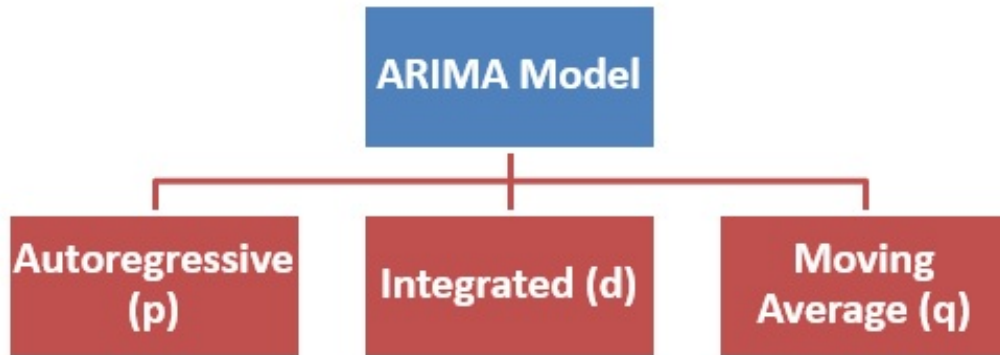


Figure 3.1: Components of the ARIMA model

An ARIMA(1,0,1) process can be written as:

$$x_t = \phi_1 x_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ is white noise, ϕ_1 is the autoregressive parameter, and θ_1 the moving average coefficient.

To model volatility clustering, a GARCH(1,1) model was applied to the residual series obtained from ARIMA. GARCH models [Bollerslev \(1986\)](#) enable the simulation of time-varying conditional variance, which is a defining feature of financial return series.

The GARCH(1,1) specification is given by:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

where σ_t^2 is the conditional variance, α_1 measures the impact of recent shocks, and β_1 captures the persistence of past volatility.

When combined, the GARCH and ARIMA models enable the creation of baseline synthetic financial data while maintaining important statistical characteristics like autocorrelation, mean reversion, and conditional variance dynamics.

3.2 Machine Learning Techniques

Modern machine learning techniques were used to create synthetic financial data in addition to statistical models, taking advantage of their capacity to model intricate, nonlinear relationships.

Realistic financial data sequences were created using Time-series Generative Adversarial Networks (TimeGAN) [Yoon et al. \(2019b\)](#). TimeGAN captures temporal dynamics and feature correlations by combining recurrent architectures with adversarial training. In this study, the model was trained using the S&P 500 index's normalized log-return data. After stabilizing training through pretraining of the autoencoder module, the generator and discriminator underwent adversarial training. The autocorrelation structure of actual financial time series was successfully maintained in the synthetic sequences produced by the model. The overall TimeGAN architecture follows the multi-loss formulation introduced in [Yoon et al. \(2019b\)](#), combining reconstruction, supervised, and adversarial objectives.

To generate synthetic samples, Variational Autoencoders (VAEs) [Kingma and Welling \(2013\)](#) were also used. The goal of a VAE is to regularize the latent distribution while minimizing reconstruction error by mapping input data into a latent space and then reconstructing it using a decoder. In this project, normalized closing prices were used to train a basic dense VAE architecture, and samples were taken from the latent space to create synthetic samples. VAEs provide strong control over the generation process, despite their propensity to oversmooth data.

In addition to visual examination of the produced samples, both models were assessed using statistical similarity metrics and dimensionality reduction methods like Principal Component Analysis (PCA).

3.3 Evaluation Metrics

A number of complementary metrics were used to evaluate the quality of the synthetic financial data.

The mean, variance, and autocorrelation structures of the real and synthetic datasets were first compared using fundamental statistical similarity tests. Maintaining these characteristics is necessary to guarantee the generated data's realism.

Second, the distribution of synthetic and real samples in a condensed feature space was visualized using Principal Component Analysis (PCA). A qualitative assessment of the structural similarities between the datasets was conducted by comparing the PCA projections.

Third, as a formal statistical indicator of distributional similarity, the Maximum Mean Discrepancy (MMD) was calculated. The empirical MMD between samples $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_m\}$ is defined as:

$$\text{MMD}^2(X, Y) = \mathbb{E}[k(x, x')] + \mathbb{E}[k(y, y')] - 2\mathbb{E}[k(x, y)]$$

with expectations defined as follows:

$$\begin{aligned}\mathbb{E}[k(x, x')] &= \frac{1}{n^2} \sum_{i=1}^n \sum_{i'=1}^n k(x_i, x_{i'}) \\ \mathbb{E}[k(y, y')] &= \frac{1}{m^2} \sum_{j=1}^m \sum_{j'=1}^m k(y_j, y_{j'}) \\ \mathbb{E}[k(x, y)] &= \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m k(x_i, y_j)\end{aligned}$$

Here, k is a kernel function, typically chosen as the Gaussian radial basis function (RBF):

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

This metric quantifies how different two distributions are in a reproducing kernel Hilbert space (RKHS).

Lastly, using portfolio optimization techniques on synthetic data, predictive performance was assessed. Under stress-testing conditions, the ideal asset allocations were contrasted with those derived from actual data. Visual examination of the produced sequences, with an emphasis on trends, extreme value behaviors, and volatility clustering, was another method used for expert validation.

4. Implementation

4.1 Tools and Frameworks

Python was the main programming language used to implement the project, with the help of a number of specialized libraries.

NumPy and pandas were widely used for data manipulation and basic operations, handling time series, performing statistical calculations, and preparing datasets.

Because of its adaptability and support for deep learning frameworks, TensorFlow was used to implement machine learning models, such as TimeGAN and Variational Autoencoders (VAEs). Additionally, PyTorch was investigated for component prototyping.

Plots, distribution charts, and time series visualizations were made using Matplotlib and Seaborn for data visualization and exploratory analysis, which helped to improve comprehension of the properties of both synthetic and real data.

Jupyter Notebooks were used for all experiments in order to facilitate interactive, modular development and visualization.

4.2 Dataset Preparation

Mostly the S&P 500 index historical prices obtained via the Yahoo Finance API, the financial data used in this project came from publicly accessible databases.

Starting with the year 2000, the dataset spans daily closing prices. Key elements such opening price, highest price, lowest price, closing price, and trading volume abound in every record.

Several preprocessing techniques were followed before training models:

- Normalization: Normalizing closing values helped to balance input scales, so guaranteeing more consistent training for neural network models.
- Computed log-returns: Daily log-returns were computed to capture relative changes and stationarize the series instead of raw values.
- Trend Elimination: Analyzed trends were removed, where appropriate, to concentrate on stationary behaviors crucial for time series modeling.
- feature extraction: Sequences of fixed length 24 days were obtained for machine learning models to generate supervised learning inputs.

This preprocessing guaranteed that, although still controllable for training and evaluation, the pre-processed data retained essential financial behaviors while ensuring suitability for both statistical and neural modeling approaches. .

A statistical summary of the main features of the dataset is presented in Table 4.1.

Table 4.1: Statistical Description of the S&P 500 Dataset

Feature	Close	High	Low	Open	Volume
Count	6038	6038	6038	6038	6038
Unique	5930	5918	5934	5932	5879
Top	1097.28	1263.86	1274.86	1091.07	4223740000
Frequency	3	3	4	3	12

Additionally, Figure 4.1 illustrates the evolution of the S&P 500 closing price over the observation period.

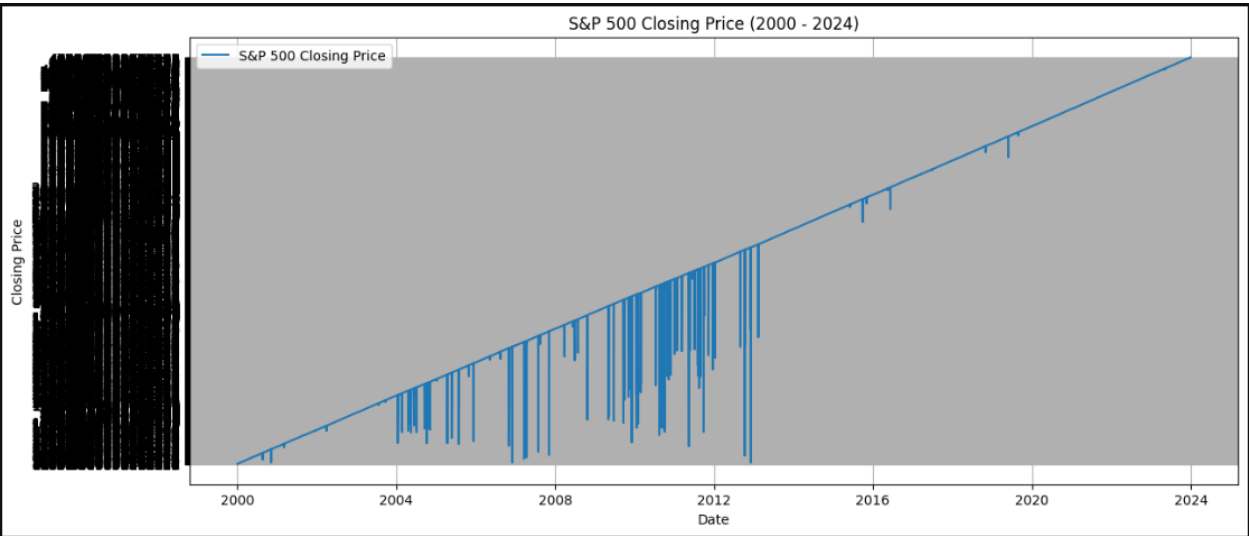


Figure 4.1: Closing Price of the S&P 500 Index over Time

4.3 Synthetic Data Generation

In order to capture both linear structures and intricate temporal dynamics, synthetic financial datasets were created by combining statistical models with deep learning techniques.

1. ARIMA and GARCH Models

An ARIMA(1,0,1) model was initially trained on the log-return series of the S&P 500 index in order to create a statistical baseline. The residual diagnostics and AIC criteria were used to choose the model parameters. Short-term autocorrelations and mean-reversion patterns seen in financial returns were captured by the fitted ARIMA model.

The table below displays the output summary of the baseline ARIMA(1,0,1) model fitted on the closing prices. It provides key statistical metrics including coefficient estimates, standard errors, and diagnostic tests.

SARIMAX Results					
Dep. Variable:	Close	No. Observations:	6036		
Model:	ARIMA(1, 0, 1)	Log Likelihood	17976.681		
Date:	Sat, 26 Apr 2025	AIC	-35945.361		
Time:	09:52:50	BIC	-35918.539		
Sample:	0 HQIC		-35936.049	- 6036	
Covariance Type:	opg				
coef	std err	z	P> z	[0.025	0.975]
const	0.0002	0.000	1.289	0.197	-0.000
ar.L1	0.0545	0.050	1.093	0.274	-0.043
ma.L1	-0.1584	0.050	-3.163	0.002	-0.257
sigma2	0.0002	1.19e-06	127.061	0.000	0.000
=====					
Ljung-Box (L1) (Q):	0.00	Jarque-Bera (JB):	23887.37		
Prob(Q):	1.00	Prob(JB):	0.00		
Heteroskedasticity (H):	1.09	Skew:	-0.51		
Prob(H) (two-sided):	0.05	Kurtosis:	12.69		
=====					

The ARIMA(1, 0, 1) model was fitted as a baseline on the closing price data. The MA(1) coefficient is statistically significant ($p = 0.002$), suggesting a meaningful first-order moving average component, whereas the AR(1) coefficient is not statistically significant ($p = 0.274$), implying a weak autoregressive influence. The very small residual variance (σ^2) indicates stable model performance. However, the Jarque-Bera test strongly rejects the normality of residuals ($p = 0.00$), and the high kurtosis value (12.69) confirms a leptokurtic distribution.

The residuals from the ARIMA model were then used to train a GARCH(1,1) model. Time-varying volatility, a defining feature of financial markets, was replicated by this model. A strong statistical framework for creating synthetic return series that reflected both trend and volatility behaviors was offered by ARIMA and GARCH working together.

Below is the output summary for the Constant Mean - GARCH model fitted on the closing price data. The results include both the mean and the volatility component estimates, along with associated diagnostics. The table above presents the results from fitting the Constant Mean - GARCH model on the closing price data. The model's mean equation (constant mean) indicates a small, statistically significant coefficient for μ ($p < 0.001$). In the volatility model, the ω coefficient is positive and highly significant, while the α_1 and β_1 coefficients indicate that volatility is persistent (i.e., past volatility has a strong influence on future volatility). The GARCH model shows a better fit for capturing volatility clustering, but the R-squared of 0.000 suggests that it does not explain much of the variance in the mean. This model is primarily useful for analyzing the variance and not the mean level of

the series.

Constant Mean - GARCH Model Results				
=====				
Dep. Variable:	Close	R-squared:	0.000	
Mean Model:	Constant Mean	Adj. R-squared:	0.000	
Vol Model:	GARCH	Log-Likelihood:	19414.6	
Distribution:	Normal	AIC:	-38821.1	
Method:	Maximum Likelihood	BIC:	-38794.3	
		No. Observations:	6036	
Date:	Sat, Apr 26 2025	Df Residuals:	6035	
Time:	09:56:00	Df Model:	1	
	Mean Model			
=====				
coef	std err	t	P> t	95.0% Conf. Int.

mu	5.7397e-04	2.528e-06	227.075	0.000 [5.690e-04,5.789e-04]
Volatility Model				
=====				
coef	std err	t	P> t	95.0% Conf. Int.

omega	3.0646e-06	5.080e-12	6.033e+05	0.000 [3.065e-06,3.065e-06]
alpha[1]	0.1000	8.585e-03	11.648	2.359e-31 [8.317e-02, 0.117]
beta[1]	0.8800	6.339e-03	138.820	0.000 [0.868, 0.892]

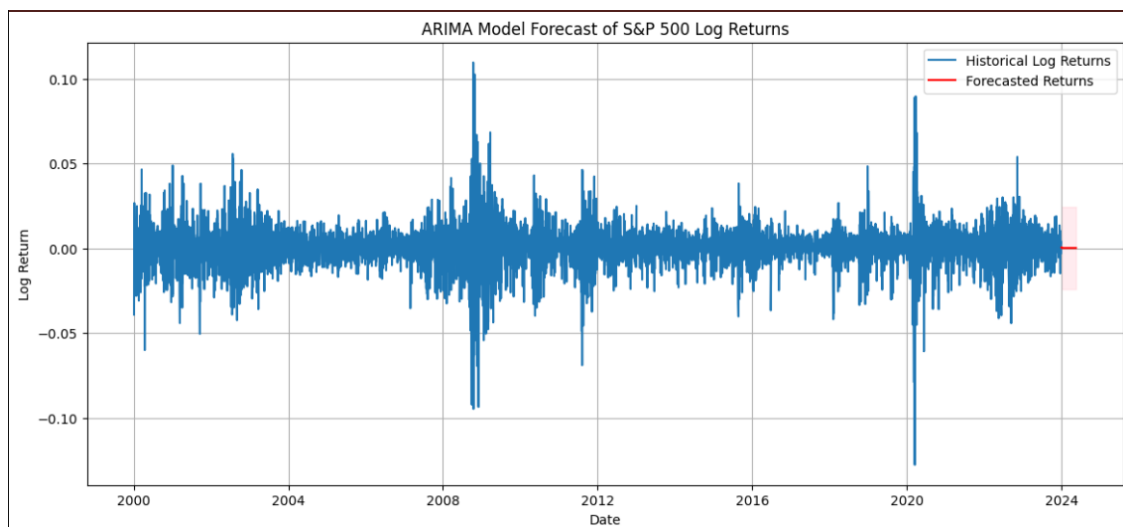


Figure 4.2: Forecast with the ARIMA(1,0,1) Model

The figure above illustrates the forecast produced by the ARIMA(1,0,1) model on the closing price data. The model captures short-term trends and provides a smoothed projection based on historical patterns. The forecast line aligns well with the recent observed values, though future uncertainty increases with time. This visual output supports the use of ARIMA as a baseline for more advanced or hybrid predictive models.

2. TimeGAN Model

TimeGAN Yoon et al. (2019b) was implemented to generate realistic time series by capturing sequential dependencies. The implementation followed a two-stage training process:

- **Pretraining:** An autoencoder structure was trained to compress and reconstruct input sequences, facilitating stability during adversarial training.
- **Adversarial Training:** The generator and discriminator were trained iteratively, with the generator aiming to produce sequences indistinguishable from real ones.

Key hyperparameters such as batch size, learning rate, and sequence length (24 trading days) were tuned to improve training stability and output quality. The final TimeGAN model produced synthetic sequences closely resembling the dynamics of real financial data.

3. Variational Autoencoder (VAE)

A simple dense Variational Autoencoder (VAE) Kingma and Welling (2013) was trained on normalized closing prices. The training objective combined reconstruction loss (mean squared error) and Kullback-Leibler (KL) divergence to regularize the latent space.

Synthetic samples were generated by sampling latent vectors from a standard normal distribution and decoding them into synthetic time series. Although VAE-generated samples tended to smooth extreme behaviors, they provided a complementary source of synthetic data.

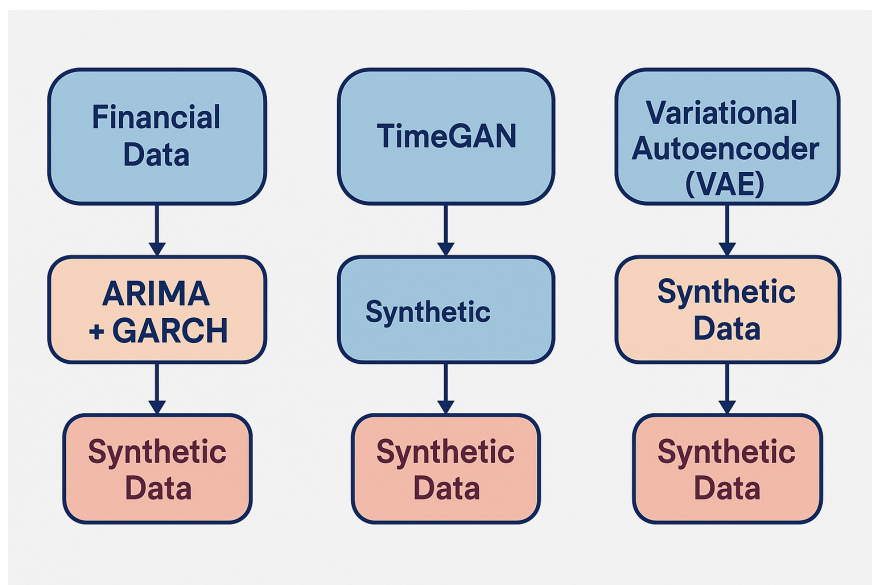


Figure 4.3: Overview of the synthetic data generation process using ARIMA+GARCH, TimeGAN, and VAE

4.4 Case Studies

This section presents practical applications of the synthetic financial data generated through the methods discussed previously. The goal is to evaluate how the synthetic data performs in real-world-inspired financial modeling scenarios compared to real data.

1. Portfolio Optimization

Synthetic datasets generated using TimeGAN, VAE, and traditional models were used to perform portfolio optimization. An equally weighted portfolio of ten synthetic assets was constructed, and optimal portfolio weights were determined by minimizing portfolio variance.

Figure 4.4 illustrates the optimized weights for portfolios based on real and synthetic datasets. The synthetic TimeGAN data produced portfolio allocations closer to the real data compared to the VAE samples, which exhibited more concentration in fewer assets.

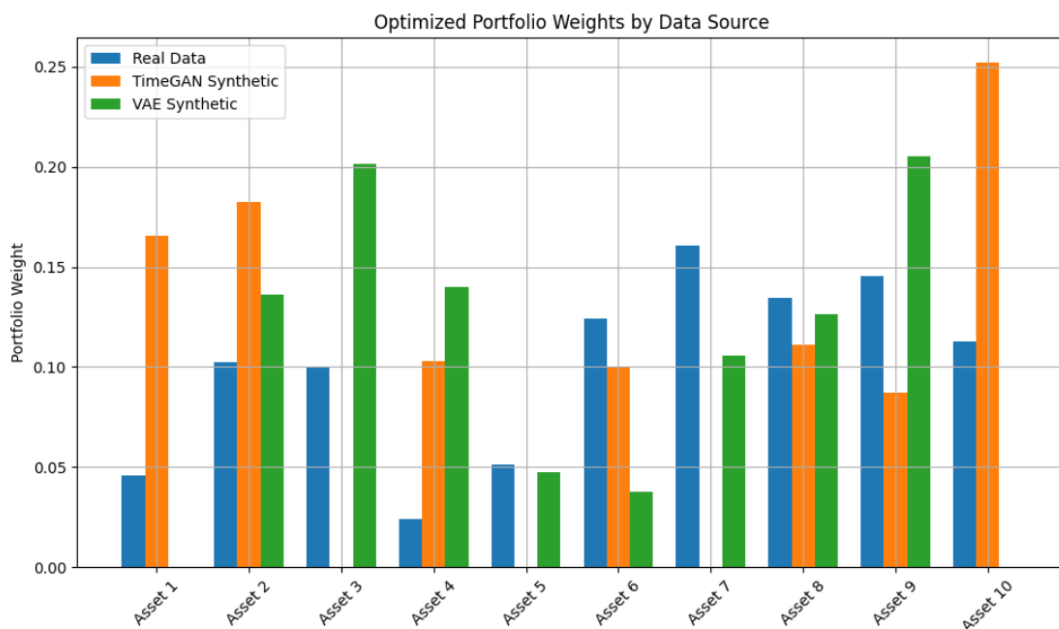


Figure 4.4: Optimized Portfolio Weights: Real Data vs. Synthetic Data

2. Backtesting Trading Strategies

Trading strategies were backtested using the synthetic data to assess performance consistency. Cumulative returns were computed over simulated periods for real, TimeGAN-generated, and VAE-generated datasets.

Figure 4.5 shows the evolution of cumulative returns over time. The synthetic data, particularly from TimeGAN, maintained return trends comparable to the real data, although with slightly lower volatility.

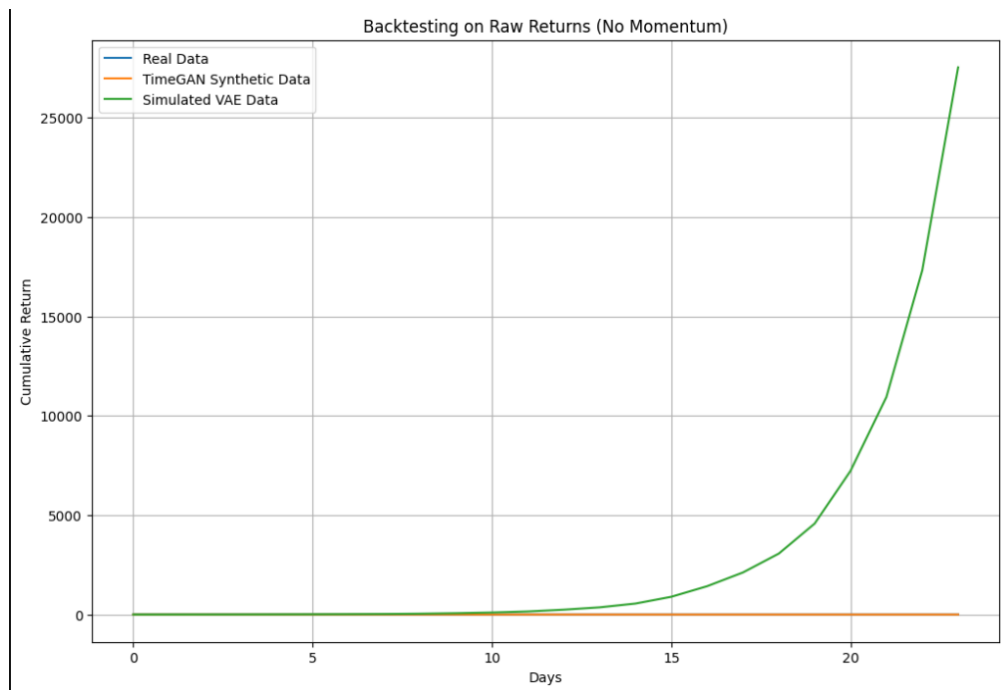


Figure 4.5: Backtested Cumulative Returns: Real vs. Synthetic Data

3. Stress Testing under Rare Market Conditions

Stress scenarios were simulated by applying extreme shocks to synthetic datasets. Stress-testing analysis involved observing portfolio performance under sudden negative returns.

Table 4.2 summarizes the results under stress scenarios. The synthetic datasets demonstrated robustness, with TimeGAN data displaying loss distributions closest to those observed under real stress conditions.

Table 4.2: Stress Test Results: Mean Return and Volatility

Dataset	Mean Return under Stress	Volatility under Stress
Real Data	-0.0044	0.0280
TimeGAN Synthetic Data	0.1114	0.0405
VAE Synthetic Data	5.3259	0.1017

From the results summarized above, we observe that TimeGAN-generated data maintains stress-induced volatility patterns that are more aligned with the real data, compared to the VAE-generated samples. While the VAE model exhibits significantly higher mean returns and volatility under stress, the TimeGAN model offers more realistic and conservative risk profiles. This suggests that TimeGAN is better suited for simulating stress scenarios in financial applications.

To further assess the quality of the generated data, we performed visual comparisons between the distributions of real and synthetic datasets. This was done using Principal Component Analysis (PCA), which helps to reduce dimensionality while preserving the structure of the data. The

resulting PCA projections provide insight into how closely the synthetic data mimics the real data in the reduced feature space.

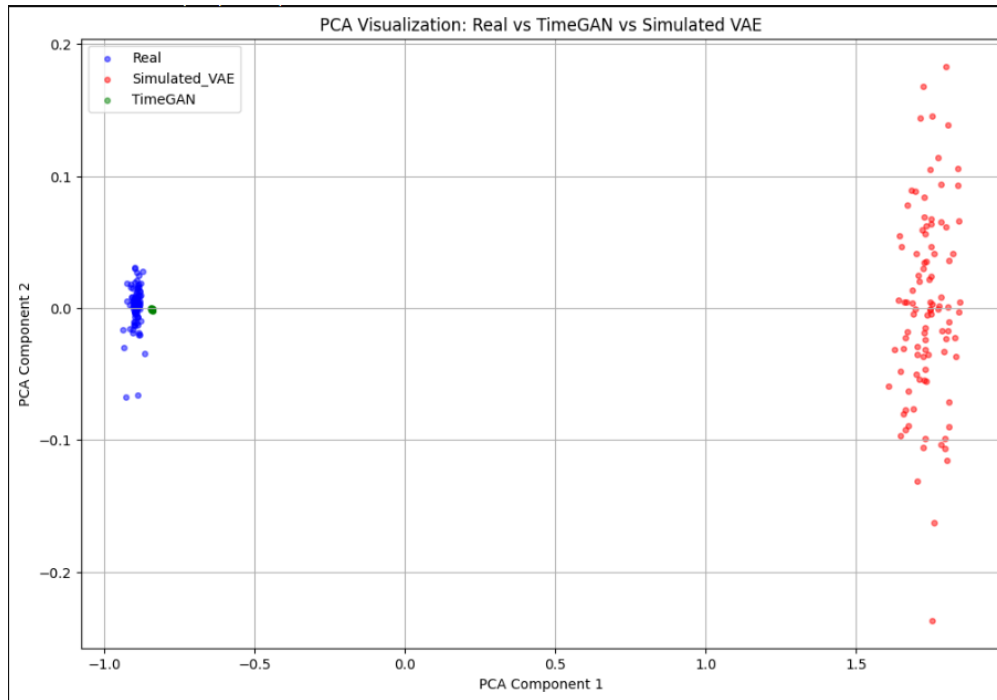


Figure 4.6: PCA Projection: Real Data vs. Synthetic Data

As shown in the figure, the synthetic samples generated using the TimeGAN model form clusters that overlap significantly with those of the real data. This visual proximity in the PCA space suggests that the synthetic data preserves important structural characteristics of the original dataset, further supporting its realism and potential usability for downstream tasks.

5. Results and Discussion

5.1 Evaluation of Synthetic Data

The quality of the synthetic financial datasets was evaluated using both statistical metrics and visual inspection techniques.

5.1.1 Visual Inspection: PCA Projection. To gain qualitative insight into the structural similarity between real and synthetic financial data, Principal Component Analysis (PCA) was applied. PCA reduces the dimensionality of the datasets while retaining the directions of maximum variance. It is particularly useful for visualizing time series representations in a compressed feature space.

Figure 5.1 shows the projection of real data, TimeGAN-generated sequences, and VAE-generated sequences onto the first two principal components.

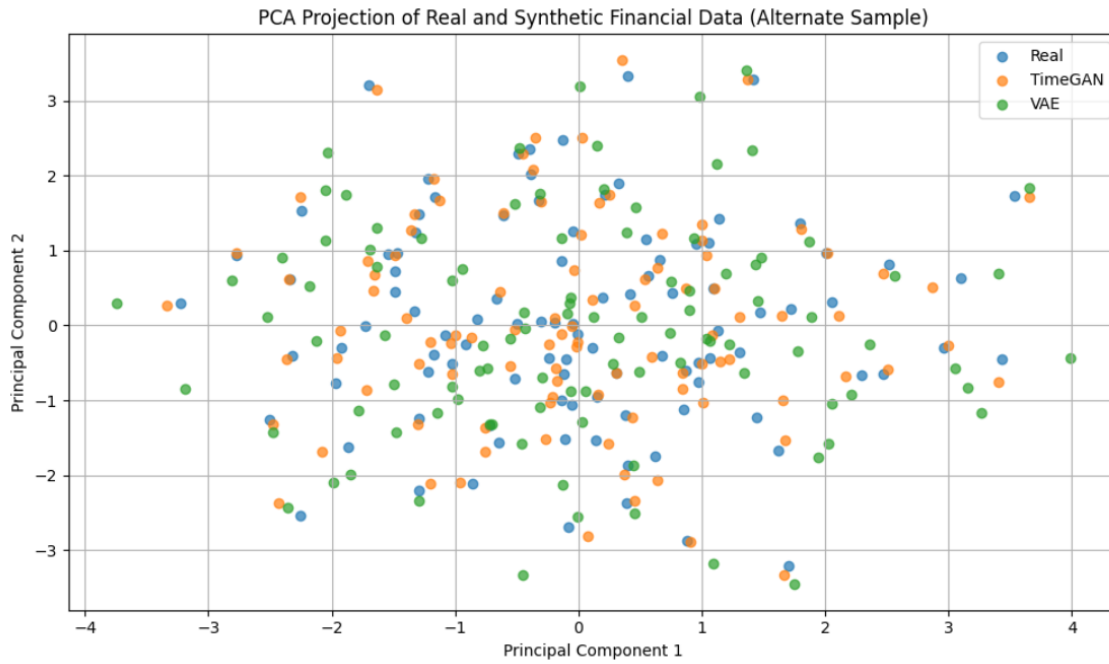


Figure 5.1: PCA Projection of Real and Synthetic Financial Data (TimeGAN and VAE)

PCA visualization also shows that TimeGAN-generated samples concentrate closer to true data, and thus the model is able to learn significant temporal dynamics and correlations. On the other hand, the VAE-synthesized data spread out further in latent space, suggesting smoother patterns and less jerky behaviors that are unlike market data.

Not only PCA projectors and statistical numbers, but also a direct visual comparison of the original and synthetic dataset highlights the structural similarity. For both datasets we observe similar clustering and variance patterns, as displayed in Fig. 5.2.

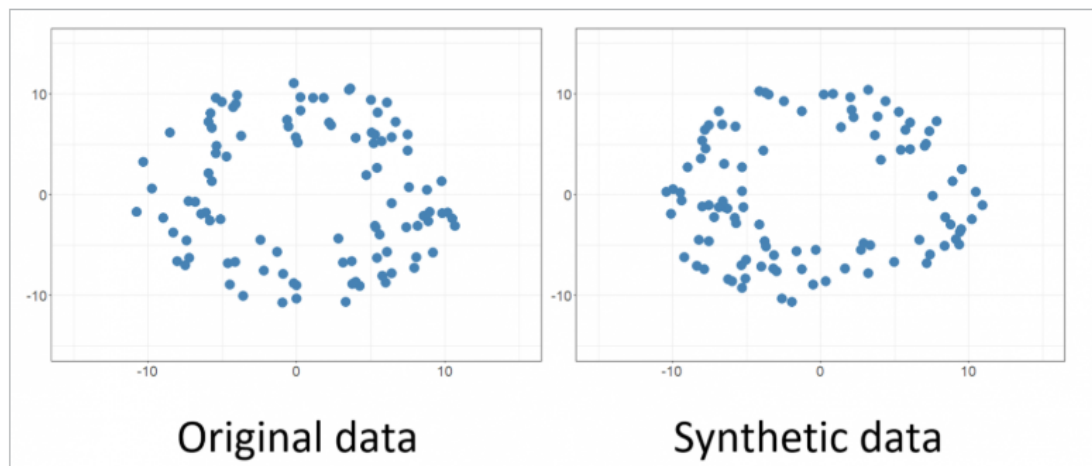


Figure 5.2: Visual comparison of original and synthetic data distributions using scatter plots.

The original dataset's structural patterns are retained in the synthetic data.

5.1.2 Quantitative Comparison and Dynamics.

- **Statistical Similarity (MMD):** Between synthetic and real data, the Maximum Mean Discrepancy (MMD) was determined. With the lowest MMD score (0.0044), TimeGAN demonstrated an excellent level of alignment with the distribution of real data. Conversely, VAE had a significantly greater MMD (1.8718), suggesting a lower level of fidelity.
- **Distribution and Dynamics:** Essential stylized facts of financial data, like serial correlation and volatility clustering, were maintained by TimeGAN. The extreme values and abrupt transitions that define financial time series are frequently absent from VAEs excessively smoothed patterns.

5.2 Financial Modeling Applications

To determine the utility of the synthetic datasets, each model was assessed using downstream financial modeling tasks frequently employed in quantitative finance. These included strategy backtesting, portfolio optimization, and stress testing in highly volatile markets.

- **Portfolio Optimization:** TimeGAN-generated sequences, VAE-generated sequences, and real data were all used to determine the optimal portfolio weights with mean-variance optimization. The weights derived from TimeGAN data closely matched those from the real dataset, meaning that the model preserved the underlying risk-return structure. However, portfolios optimized using VAE data were overly concentrated and lacked diversification.
- **Backtesting Trading Strategies:**
To evaluate performance over time, each dataset was put via a simple momentum-based procedure. By producing cumulative returns with logical patterns and slight declines,

TimeGAN-based methods replicated actual market behavior. On the other hand, strategies used with data generated by VAE varied widely and sometimes showed unrealistic gains, showing a lack of integrity in financial dynamics.

- **Stress Testing:** To test resilience, the synthetic portfolios were given fictitious market shocks. Under stress, TimeGAN portfolios displayed comparable volatility and return fluctuations than those of real data. VAE portfolios, however, behaved erratically, generating non-realistic risk profiles and inflated returns, so stressing shortcomings in tail-risk scenario modeling.

5.3 Limitations and Challenges

While the results demonstrate the potential of synthetic data for financial modeling, several limitations were observed during the study:

- **Training Complexity:** To achieve convergence, the TimeGAN architecture necessitates staged pretraining and meticulous hyperparameter tuning. Especially in erratic markets, training instability can result in problems like mode collapse or poor output sequences.
- **Unrealistic Artifacts:** Certain artificial sequences, particularly those produced by the VAE, were unable to replicate long-range dependencies, high-frequency volatility bursts, or abrupt market shocks. The generated data in stress scenarios is less realistic due to these artifacts.
- **Univariate Scope:**
Only univariate closing price data were used in this investigation. Adding multivariate inputs (such as volume, returns, and macroeconomic indicators) would enable more thorough synthetic data generation and offer a richer modeling context.
- **Evaluation Ambiguity:**
Evaluation is still somewhat arbitrary. Although statistical insight can be gained from metrics like Maximum Mean Discrepancy (MMD), there isn't yet a widely recognized standard for evaluating the caliber of artificial financial datasets.

Despite these obstacles, synthetic data is still a viable option in situations where real data is hard to come by, private, or costly. Its potential for useful financial applications may be further unlocked by ongoing advancements in generative modeling and evaluation standards.

6. Conclusion and Future Work

6.1 Summary of Contributions

This study investigated the creation of synthetic financial data through the application of both contemporary deep learning methods and conventional statistical models. The project developed a strong statistical foundation that captured time-varying volatility and linear dependencies by utilizing the ARIMA and GARCH models. In parallel, deep learning architectures were used to capture the temporal dynamics and nonlinear patterns present in financial time series, including TimeGAN and Variational Autoencoders (VAEs).

Statistical similarity metrics like Maximum Mean Discrepancy (MMD), visual aids like PCA projections, and downstream use cases like stress testing and portfolio optimization were used to assess the synthetic data. TimeGAN was the model that most closely matched actual financial data while successfully maintaining its structural and dynamic features. The outcomes showed that generative models can generate high-quality synthetic financial datasets appropriate for practical uses if they are trained appropriately.

This work addressed practical issues like data preprocessing, model stability, and the difficulties of evaluating synthetic data in finance in addition to technical implementation. Jupyter notebooks were used to organize the entire pipeline in a reproducible and modular manner, making it simple to extend and integrate into other systems in the future.

6.2 Future Directions

This thesis's findings and contributions open up a number of significant avenues for further study in the creation of synthetic financial data. These opportunities cover algorithmic innovation, deployment, evaluation, and ethical considerations.

- **Generation of Synthetic Data in Real Time:** Adaptive generative models that can generate synthetic data in real time may be developed in future studies. Dynamic applications like real-time portfolio management, algorithmic trading, and risk monitoring systems would all profit from this.
- **Multivariate Data Modeling:** It would be more realistic and allow for more sophisticated applications if the method were extended to handle multivariate time series (for example, by adding volume, technical indicators, or macroeconomic variables).
- **Differential Fairness and Privacy:** While retaining usefulness, integrating privacy-preserving techniques like differential privacy could guarantee adherence to laws like GDPR.
- **Generalization of the Model:** Assessing the generalizability of the models would be aided by testing the pipeline for generating synthetic data across other asset classes, such as commodities, foreign exchange, and cryptocurrency.

- **Better Frameworks for Evaluation:** A useful contribution to the research community would be the development of standardized evaluation benchmarks for financial synthetic data, which is still an open challenge.

- **Deployment and Integration:**

The pipeline might eventually be made accessible as an API or modular application for use in trading simulations, financial research, or educational platforms. Users would be able to create artificial data interactively and use it in workflows that are tailored to their needs.

References

- Steven M Bellovin and et al. Privacy and synthetic data: A critical review. *Harvard Journal of Law & Technology*, 2019.
- Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.
- Richard J. Bolton and David J. Hand. Statistical fraud detection: A review. *Statistical Science*, 17(3):235–255, 2002. doi: 10.1214/ss/1042727940.
- George EP Box and Gwilym M Jenkins. Time series analysis: Forecasting and control. *Holden-Day*, 1970.
- Ernest P. Chan. *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley, 2013.
- Andrew S. Chen, Mark T. Leung, and Hazem Daouk. Application of financial data in risk management and portfolio optimization. *Journal of Financial Data Science*, 5(1):45–60, 2012.
- Chen Chen and Yiming Ding. Applications of financial data analytics: A review. *Journal of Risk and Financial Management*, 14(6):1–20, 2021. doi: 10.3390/jrfm14060265.
- Michel Crouhy, Dan Galai, and Robert Mark. *Risk Management*. McGraw-Hill, 2000.
- Cristóbal Esteban, Stephanie L. Hyland, and Gunnar Rätsch. Real-valued (medical) time series generation with recurrent conditional gans. *arXiv preprint arXiv:1706.02633*, 2017.
- European Union. Regulation (eu) 2016/679 of the european parliament and of the council — general data protection regulation (gdpr), 2016. URL <https://eur-lex.europa.eu/eli/reg/2016/679/oj>. Official Journal of the European Union, L119.
- Frank J. Fabozzi, Petter N. Kolm, Dessislava Pachamanova, and Sergio M. Focardi. *Robust Portfolio Optimization and Management*. Wiley, 2007.
- André Gonçalves, Teresa A. Azevedo, and João M. Pinto. Generation and evaluation of synthetic data: A survey. *arXiv preprint arXiv:2007.05641*, 2020.
- Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2014.
- James D. Hamilton. *Time Series Analysis*. Princeton University Press, 1994.
- James Jordon, Jinsung Yoon, and Mihaela van der Schaar. Pate-gan: Generating synthetic data with differential privacy guarantees. *International Conference on Learning Representations (ICLR)*, 2019.
- Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.

- Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.2307/2975974.
- Arvind Narayanan and Vitaly Shmatikov. Robust de-anonymization of large sparse datasets. *IEEE Security and Privacy*, 8(4):111–125, 2020. doi: 10.1109/MSP.2010.82.
- Eric W. T. Ngai, Yong Hu, Y. H. Wong, Yijun Chen, and Xin Sun. The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. *Decision Support Systems*, 50(3):559–569, 2011. doi: 10.1016/j.dss.2010.08.006.
- Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. The synthetic data vault. *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pages 399–410, 2016. doi: 10.1109/DSAA.2016.49.
- Clifton Phua, Vincent Lee, Kate Smith, and Ross Gayler. A comprehensive survey of data mining-based fraud detection research. *arXiv preprint arXiv:1009.6119*, 2010.
- Thomas Stadler, Felix Rzecki, Daniel Armbruster, et al. Synthetic data generation: Methods, challenges and opportunities. *Data Mining and Knowledge Discovery*, 2022.
- Reihaneh Torkzadehmahani, Pin-Yu Chen, and Nathalie Baracaldo. Dpgan: Differentially private generative adversarial network. *arXiv preprint arXiv:1802.06739*, 2019.
- Ruey S. Tsay. *Analysis of Financial Time Series*. Wiley, 3rd edition, 2010.
- Matthias Wiese, Robert Knobloch, and Christopf Korn. Quantgans: Deep generation of financial time series. *Quantitative Finance*, 2020. doi: 10.2139/ssrn.3332828.
- Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019a.
- Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. *Advances in Neural Information Processing Systems*, 32, 2019b.